1. Introduction & Features

1. **C++ is primarily an extension of which language?**
   a) Java
   b) Python
   c) C
   d) Fortran
   ✅ **Answer: c) C**
2. **Which of the following are features of C++? (Select all correct)**
   a) Object-Oriented
   b) Portable
   c) Interpreted
   d) Rich Library
   e) Platform Dependent
   ✅ **Answer: a, b, d**
3. **C++ supports which type(s) of programming?**
   a) Procedural only
   b) Object-Oriented only
   c) Both Procedural and Object-Oriented
   d) Functional only
   ✅ **Answer: c) Both Procedural and Object-Oriented**

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2. History

4. **Who developed C++?**
   a) Dennis Ritchie
   b) Bjarne Stroustrup
   c) James Gosling
   d) Guido van Rossum
   ✅ **Answer: b) Bjarne Stroustrup**
5. **C++ was originally called:**
   a) C+
   b) C with Classes
   c) Objective C

d) C#
✅ **Answer: b) C with Classes**

6. **The first ISO standard for C++ was released in:**
   a) 1985
   b) 1990
   c) 1998
   d) 2003
   ✅ **Answer: c) 1998**

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3. Program Structure

7. **Every C++ program must have:**
   a) A class
   b) A main() function
   c) A constructor
   d) A destructor
   ✅ **Answer: b) A main() function**
8. **Which comes first in a C++ program?**
   a) main() function
   b) Preprocessor directives
   c) Variable declarations
   d) Comments
   ✅ **Answer: b) Preprocessor directives**
9. **What is the correct order of program structure?**
   a) main(), #include, statements, comments
   b) #include, main(), declarations, statements
   c) Statements, main(), #include, comments
   d) Comments, statements, #include, main()
   ✅ **Answer: b) #include, main(), declarations, statements**

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4. Tokens

10. **Which of these are tokens in C++? (Select all correct)**
    a) Keywords

b) Identifiers
c) Comments
d) Operators
e) Punctuators
✅ **Answer: a, b, d, e**

11. **The smallest meaningful unit in a C++ program is called:**
    a) Statement
    b) Token
    c) Expression
    d) Block
    ✅ **Answer: b) Token**

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5. Keywords & Identifiers

12. **Which of these is a valid identifier?**
    a) 1number
    b) _count
    c) total marks
    d) int
    ✅ **Answer: b) _count**
13. **Which rule applies to identifiers?**
    a) Must start with a digit
    b) Can contain spaces
    c) Case-sensitive
    d) Can be a keyword
    ✅ **Answer: c) Case-sensitive**

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6. Data Types & Variables

14. **Which data type is used for decimal numbers?**
    a) int
    b) float
    c) char

d) bool
✅ **Answer: b) float**

15. **A variable is:**
    a) A reserved word
    b) A named memory location
    c) A function
    d) An operator
    ✅ **Answer: b) A named memory location**
16. **Which is the correct variable declaration?**
    a) int 1x;
    b) float mark;
    c) char name[];
    d) double 3.14;
    ✅ **Answer: b) float mark;**

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7. Input/Output

17. **For output in C++, we use:**
    a) cin
    b) cout
    c) printf
    d) scanf
    ✅ **Answer: b) cout**
18. **The extraction operator is:**
    a) <<
    b) >>
    c) ->
    d) ::
    ✅ **Answer: b) >>**
19. **Which statement reads an integer from the user?**
    a) cin << x;
    b) cout >> x;
    c) cin >> x;
    d) cout << x;
    ✅ **Answer: c) cin >> x;**

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8. Operators & Expressions

20. **Which operator is used for remainder?**
    a) /
    b) %
    c) *
    d) &
    ✅ **Answer: b) %**
21. **An expression is:**
    a) A complete instruction
    b) A combination of operators and operands
    c) A data type
    d) A function call
    ✅ **Answer: b) A combination of operators and operands**
22. **What is the result of 5 + 3 * 2?**
    a) 16
    b) 11
    c) 13
    d) 10
    ✅ **Answer: b) 11**

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9. Conditional & Looping Statements

23. **Which loop executes at least once?**
    a) for
    b) while
    c) do-while
    d) if-else
    ✅ **Answer: c) do-while**
24. **The break statement is used to:**
    a) Skip an iteration
    b) Exit a loop
    c) Continue to next iteration
    d) Stop the program
    ✅ **Answer: b) Exit a loop**
25. **Which is a valid if statement?**
    a) if x > 5

b) if (x > 5)
c) if {x > 5}
d) if [x > 5]
✅ **Answer: b) if (x > 5)**

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10. Functions

26. **A function can return:**
    a) One value
    b) Multiple values
    c) No value (void)
    d) Both a and c
    ✅ **Answer: d) Both a and c**
27. **Function overloading means:**
    a) Same name, different parameters
    b) Different name, same parameters
    c) Same name, same parameters
    d) Different return types only
    ✅ **Answer: a) Same name, different parameters**
28. **Inline functions are used to:**
    a) Reduce memory usage
    b) Increase execution speed
    c) Make code readable
    d) Handle errors
    ✅ **Answer: b) Increase execution speed**

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11. Arrays & Strings

29. **Array index starts from:**
    a) 0
    b) 1
    c) -1
    d) Any number
    ✅ **Answer: a) 0**

30. **Which is a valid array declaration?**
    a) int arr[5];
    b) array int[5];
    c) int arr = {1,2,3};
    d) arr[5] int;
    ✅ **Answer: a) int arr[5];**
31. **A string in C++ is:**
    a) An array of characters
    b) A class in STL
    c) A keyword
    d) Both a and b
    ✅ **Answer: d) Both a and b**

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12. Pointers & Memory

32. **A pointer stores:**
    a) A value
    b) An address
    c) A data type
    d) An operator
    ✅ **Answer: b) An address**
33. **The & operator is used to:**
    a) Get value at address
    b) Get address of variable
    c) Declare pointer
    d) Allocate memory
    ✅ **Answer: b) Get address of variable**
34. **Dynamic memory allocation uses:**
    a) malloc() and free()
    b) new and delete
    c) alloc() and dealloc()
    d) Both a and b
    ✅ **Answer: b) new and delete**

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13. Structures & Classes

35. **The default access specifier in a struct is:**
    a) private
    b) public
    c) protected
    d) static
    ✅ **Answer: b) public**
36. **A class is:**
    a) A data type
    b) A blueprint for objects
    c) A variable
    d) A function
    ✅ **Answer: b) A blueprint for objects**
37. **Which operator accesses structure members?**
    a) . (dot)
    b) -> (arrow)
    c) :: (scope)
    d) Both a and b
    ✅ **Answer: d) Both a and b**

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14. OOP Concepts

38. **Inheritance allows:**
    a) Data hiding
    b) Code reusability
    c) Polymorphism
    d) Encapsulation
    ✅ **Answer: b) Code reusability**
39. **Polymorphism means:**
    a) One name, many forms
    b) Data hiding
    c) Wrapping data
    d) Code duplication
    ✅ **Answer: a) One name, many forms**
40. **Encapsulation is:**
    a) Hiding data

b) Showing only essential details
c) Wrapping data and functions
d) Inheriting properties
✅ **Answer: c) Wrapping data and functions**

41. **A virtual function is used for:**
    a) Compile-time polymorphism
    b) Runtime polymorphism
    c) Memory management
    d) Error handling
    ✅ **Answer: b) Runtime polymorphism**

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15. Advanced Topics

42. **STL stands for:**
    a) Standard Template Library
    b) System Template Library
    c) Standard Type Library
    d) Simple Template Library
    ✅ **Answer: a) Standard Template Library**
43. **Namespace is used to:**
    a) Avoid name conflicts
    b) Increase speed
    c) Reduce memory
    d) Handle errors
    ✅ **Answer: a) Avoid name conflicts**
44. **A friend function can access:**
    a) Only public members
    b) Only private members
    c) Both public and private members
    d) Only protected members
    ✅ **Answer: c) Both public and private members**
45. **const keyword makes a variable:**
    a) Unchangeable
    b) Visible everywhere
    c) Invisible
    d) Dynamic
    ✅ **Answer: a) Unchangeable**

46. **Exception handling uses:**
    a) try, catch, throw
    b) if, else, switch
    c) for, while, do
    d) new, delete, malloc
    ✅ **Answer: a) try, catch, throw**
47. **Templates allow:**
    a) Writing generic code
    b) Faster execution
    c) Memory optimization
    d) Error prevention
    ✅ **Answer: a) Writing generic code**
48. **File handling in C++ uses which header?**
    a) `<iostream>`
    b) `<fstream>`
    c) <stdio.h>
    d) <file.h>
    ✅ **Answer: b) `<fstream>`**
49. **Which is a jump statement? (Select all correct)**
    a) break
    b) continue
    c) return
    d) goto
    e) switch
    ✅ **Answer: a, b, c, d**
50. **Operator overloading allows operators to work with:**
    a) Built-in types only
    b) User-defined types only
    c) Both built-in and user-defined types
    d) None
    ✅ **Answer: c) Both built-in and user-defined type**